ECE 310 Fall 2026
Lecture 1: Introduction to Digital Signals

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Welcome to ECE 310!

Let's take a look at the course website.



ECE 310 Frequently Asked Questions



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- Is ECE 310 harder than ECE 210?



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- Why should I care about ECE 310?



ECE 310 Frequently Asked Questions

- Is ECE 310 harder than ECE 210?
- I had a really tough time with ECE 210. Should I be concerned taking ECE 310?
- Why should I care about ECE 310?
- What questions do you have?



What is signal processing?

What are some words, phrases, or applications you associate with signal processing?



Learning Objectives

- Describe what defines a signal.
- Explain the difference between continuous-time and discrete-time signals.
- Understand the advantages of working with digital signals over analog signals.



What is a signal?



What is a signal?

- A signal is any measurable quantity that varies over time, space, or any other variable that orders or organizes these measurements.



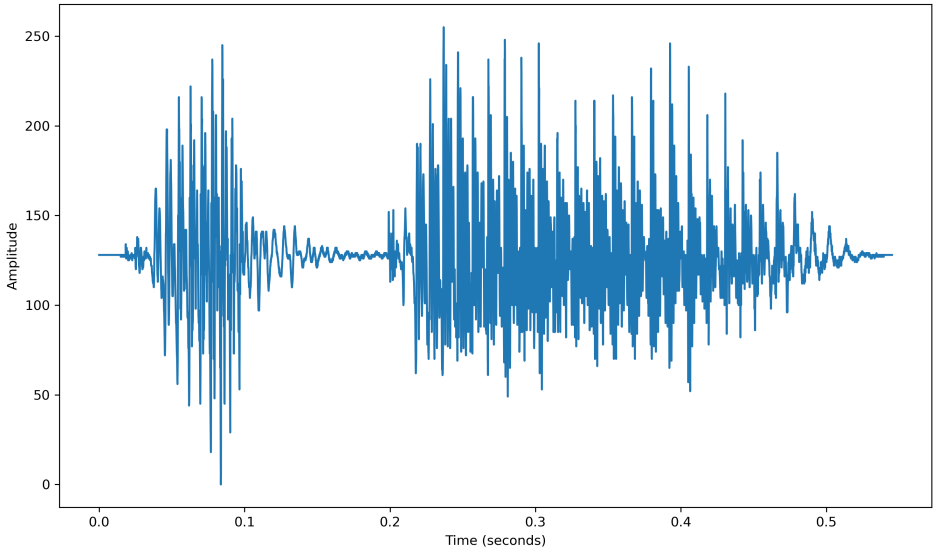
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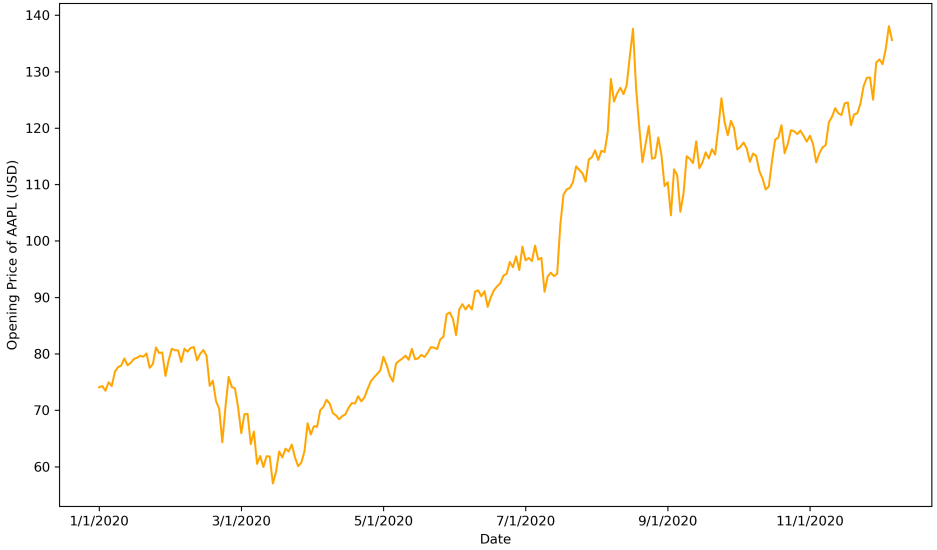
This sounds quite broad, but that's because signal processing is so widely applicable!



Signal Examples: Audio Signal



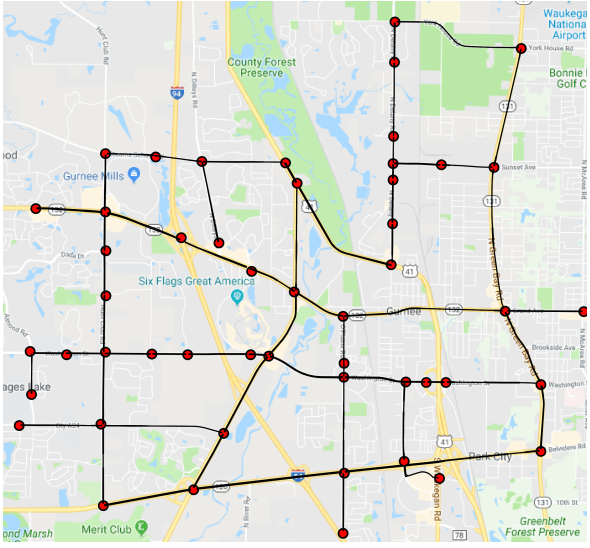
Signal Examples: Financial Signal



Signal Examples: Image Signal



Signal Examples: Graph Signal



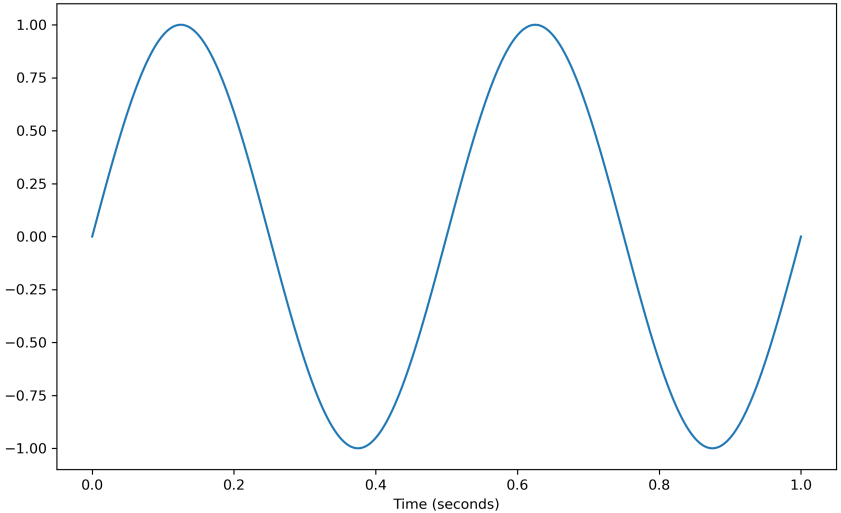
Continuous-time vs. Discrete-time Signals

- A *continuous-time* signal $x(t)$ is observed continuously over time. Every time interval contains infinitely many values from the signal.
- A *discrete-time* signal $x[n]$ is observed at regularly spaced points in time. We say that the time-axis is *discretized*.
- For example, audio is commonly recorded at 44.1 kHz. Thus, we have a new sample every $T = 1/44{,}100$ seconds. T is called the *sampling period*.

$$x[n] = x(nT)$$



Continuous-time vs. Discrete-time Signals



Continuous-valued vs. Discrete-valued Signals

- A *continuous-valued* signal takes any possible continuous value for its measurements.
- A *discrete-valued* signal takes one of a finite number of pre-defined values. The measurements are said to be *quantized*.



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- A continuous-time, continuous-valued signal is referred to as an *analog signal*.
- A discrete-time, discrete-valued signal is referred to as a *digital signal*.*



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- A continuous-time, continuous-valued signal is referred to as an *analog signal*.
- A discrete-time, discrete-valued signal is referred to as a *digital signal*.*
  - We will assume our signals are continuous-valued in this class since modern computing systems have very high precision.



Why digital signals?

What are some advantages of using digital signals over analog signals?

